

Unit 1: Logic and Proof methods

Discrete Structure



logic



Introduction:

- *It is the science dealing with the method of reasoning the mathematical models.*
- *needs some rules associated with logic so that we can apply those rules for mathematical reasoning.*
- *It is defined as the representation language of knowledge.*
- *It has practical applications to the design of computing machines, to artificial intelligence, to computer programming, and to other areas of computer science.*
- ***In conclusion, logic is the basis of reasoning.***
- *On the basis of its representation capability it is of two types:*
 - a) propositional logic*
 - b) predicate logic*



Propositional Logic:

- The area of logic which deals with proposition is known as propositional logic.
- A proposition is a declarative sentence (that is, a sentence that declares a fact) that is either true or false, but not both.
- The truth value of a proposition is true, denoted by T, if it is a true proposition, and the truth value of a proposition is false, denoted by F, if it is a false proposition.
- All the following declarative sentences are propositions.
 1. Washington, D.C., is the capital of the United States of America.
 2. Toronto is the capital of Canada.
 3. $1 + 1 = 2$.
 4. $2 + 2 = 3$.
- Propositions 1 and 3 are true, whereas 2 and 4 are false.



Propositional logic is of two types

- a) Simple propositional logic
 - b) Compound propositional logic
- a) Simple:** It express atomic propositions about the world. Atomic sentence are either propositional symbols $P, Q, R, S, \dots$ Or logical constant (true, false)
- b) Compound:** It can be formed by combining atomic formulas with the following connectives not($\neg$), and($\wedge$), or($\vee$), implication($\rightarrow$) and bi-conditional($\leftrightarrow$).

Logical Operators/Connectives



- Logical operators are used to construct mathematical statements having one or more propositions by combining the propositions.

Negation (not)

- Let p be a proposition. The negation of p , denoted by $\neg p$ (also denoted by p'), is the statement “It is not the case that p . ”
- The proposition $\neg p$ is read as “not p . ” The truth value of the negation of p , $\neg p$, is the opposite of the truth value of p .



Find the negation of the proposition “Michael’s PC runs Linux” and express this in simple English.

Solution:

The negation is

“It is not the case that Michael’s PC runs Linux.”

This negation can be more simply expressed as

“Michael’s PC does not run Linux.”

Conjunction (and):

Let p and q be propositions. The conjunction of p and q , denoted by $p \wedge q$, is the proposition “ p and q .”

The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.



Example:

If we have propositions p = “Ram is intelligent” and q = “Ram is diligent” the conjunction of p and q is Ram is intelligent and diligent. This proposition is true only when Ram is intelligent and he is diligent also, false otherwise.

Disjunction (or):

- Let p and q be propositions. The disjunction of p and q , denoted by $p \vee q$, is the proposition “ p or q .” The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.



Example:

If we have propositions $p = \text{“Ram is intelligent”}$ and $q = \text{“Ram is diligent”}$ the disjunction of p and q is

Ram is intelligent or he is diligent. This proposition is false only when Ram is not intelligent and not diligent, true otherwise.

Logical Operators/Connectives

Implication:



- Given two propositions p and q , the proposition implication $p \rightarrow q$ is the proposition that is false when p is true and q is false, true otherwise.
- Here p is called “hypothesis” or “antecedent” or “premise” and q is called “conclusion” or “consequence”.
- We come across the implication in many places in mathematical reasoning and we use different terminologies to express $p \rightarrow q$ like:
 - “if p , then q ”,
 - “ q is consequence of p ”,
 - “ p is sufficient for q ”,
 - “ q if p ”
 - “ q is necessary for p ”,
 - “ q follows from p ”,
 - “if p , q ”,
 - “ p implies q ”,
 - “ p only if q ”,
 - “ q whenever p ”,
 - “ q provides p ”
 - “ q when p ”

Logical Operators/Connectives



Example: p = “today is Sunday” q = “it is hot” then the implication can be “if today is Sunday then it is hot today” or “today is Sunday only if it is hot today”.

Truth Table:

$p \rightarrow q.$		
p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T



Contra-positive, Inverse and Converse:

Some of the related implications formed from $p \rightarrow q$ are:

- **Converse:** $q \rightarrow p$ (if it is hot today then today is Sunday).
- **Inverse:** $\neg p \rightarrow \neg q$ (if today is not Sunday then it is not hot today).
- **Contra-positive:** $\neg q \rightarrow \neg p$ (if it is not hot then today is not Sunday).



Example2:

What are the contrapositive, the converse, and the inverse of the conditional statement “the home town wins whenever it is raining?”

Soln:

because “q whenever p” is one of the ways to express the conditional statement $p \rightarrow q$ the original statement can be rewritten as
“ if it is raining, then the home team wins”

The contrapositive of this conditional statement is,
“ if the home team does not win, then it is not raining”

The converse is,
“ if the home team wins, then it is raining”

The inverse is,
“ if it is not raining, then the home team does not win”



Bi-conditional:

Given propositions p and q , the bi-conditional $p \leftrightarrow q$ is a proposition that is true when p and q have same truth values. Alternatively $p \leftrightarrow q$ is true whenever both $p \rightarrow q$ and $q \rightarrow p$ are true.

Some of the terminologies used for bi-conditional are:

- “ p if and only if q ”
- “if p then q , and conversely”
- “ p is necessary and sufficient for q ”

Example: For propositions given above in implication, “today is Sunday if and only if it is hot today”.



Truth table:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Precedence of Logical Operators



- We can construct compound propositions using the negation operator and the logical operators defined so far.
- We will generally use parentheses to specify the order in which logical operators in a compound proposition are to be applied.
- For instance, $(p \vee q) \wedge (\neg r)$ is the conjunction of $p \vee q$ and $\neg r$.
- The precedence levels of the logical operators, $\neg$, $\wedge$, $\vee$, $\rightarrow$, and $\leftrightarrow$.

Logic and Bit Operations



- A bit is a symbol with two possible values, namely, 0 (zero) and 1 (one).
- A bit can be used to represent a truth value.
- we will use a 1 bit to represent true and a 0 bit to represent false.
- Find the bitwise *OR*, *bitwise AND*, and *bitwise XOR* of the bit strings *01 1011 0110* and *11 0001 1101*.

01 1011 0110

11 0001 1101

11 1011 1111 bitwise *OR*

01 0001 0100 bitwise *AND*

10 1010 1011 bitwise *XOR*



More Examples logical connectives:

1) Let, p = “it rained last night”

q = “the sprinkles came on last night”

r = “the lawn was wet this morning”

Translate the following into English $\neg p$, $r \wedge \neg p$ and $\neg r \vee p \vee q$.

- $\neg p$ = “ it didn’t rain last night”
- $r \wedge \neg p$ = “the lawn was wet this morning and it didn’t rain last night”
- $\neg r \vee p \vee q$ = “either the lawn was not wet this morning or it rained last night or the sprinkles came on last night”

Translating English Sentences



Representing the English sentence in propositional logic:

- Restate the given sentence into building block sentences i.e. into atomic sentence.
- Give the symbol to each sentence and substitute the symbols using connectives.

For e.g. “if it is snowing then I will go to the beach”

Restate into “it is snowing” give it symbol p and “I will go to the beach” and give it symbol q then we can write it as

$$p \rightarrow q.$$

Translating English Sentences



Example1:

How can this English sentence be translated into a logical expression?

“You can access the Internet from campus only if you are a computer science major or you are not a freshman.”

Solution: we will use propositional variables to represent each sentence part and determine the appropriate logical connectives between them.

let a , c , and f represent “You can access the Internet from campus,” “You are a computer science major,” and “You are a freshman,” respectively. Noting that “only if” is one way a conditional statement can be expressed, this sentence can be represented as:

$$a \rightarrow (c \vee \neg f).$$

Translating English Sentences



Example2:

How can this English sentence be translated into a logical expression?

“You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old.”

Solution:

Let q , r , and s represent “You can ride the roller coaster,” “You are under 4 feet tall ,” and “You are older than 16 years old,” respectively.

Then the sentence can be translated to

$$(r \wedge \neg s) \rightarrow \neg q.$$



Example3:

Express the specification “ the automated reply cannot be sent when the file system is full” using logical connectives.

Soln:

Let p denote “ the automated reply can be sent”

q denote “ the file system is full”

Then, $\neg p$ represent “ *the automated reply cannot be sent*”

Now,

the conditional statement will be

$$q \rightarrow \neg p$$



Tautology, Contradiction and Contingency:

- A compound proposition that is always true, no matter what the truth values of the atomic propositions that contain in it, is called a tautology. For e.g. $p \vee \neg p$ is always true.
- A compound proposition that is always false is called contradiction. For e.g. $p \wedge \neg p$ is always false
- A compound proposition that is neither a tautology nor a contradiction is called a contingency. For e.g. $p \rightarrow q$

Logical Equivalences



Logical Equivalences:

If two propositions are semantically identical then we say those two propositions are “equivalent”.

The compound propositions p and q are logically equivalent, denoted by $p \leftrightarrow q$ or $p \equiv q$,

- If proposition $p \leftrightarrow q$ is a tautology.
- The truth table of each statement have the same truth values.

Some important logical equivalences

$$p \wedge \mathbf{T} \Leftrightarrow p$$

Identity law

$$p \vee \mathbf{F} \Leftrightarrow p$$

Identity law

$$p \wedge \mathbf{F} \Leftrightarrow \mathbf{F}$$

Domination law

$$p \vee \mathbf{T} \Leftrightarrow \mathbf{T}$$

Domination law

$$p \wedge p \Leftrightarrow p$$

Idempotent law

$$p \vee p \Leftrightarrow p$$

Idempotent law

$$\neg(\neg p) \Leftrightarrow p$$

Double negation law

$$p \wedge q \Leftrightarrow q \wedge p$$

Commutative law

$$p \vee q \Leftrightarrow q \vee p$$

Commutative law

$$(p \wedge q) \wedge r \Leftrightarrow p \wedge (q \wedge r)$$

Associative law

$$(p \vee q) \vee r \Leftrightarrow p \vee (q \vee r)$$

Associative law

$$p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$$

Distributive law

$$p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$$

Distributive law

$$\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$$

De Morgan's law

$$\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$$

De Morgan's law

$$p \wedge \neg p \Leftrightarrow \mathbf{F}$$

Trivial tautology

$$p \vee \neg p \Leftrightarrow \mathbf{T}$$

Trivial tautology

$$p \rightarrow q \Leftrightarrow \neg p \vee q$$

Implication

$$p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$$

Equivalence

$$(p \wedge q) \rightarrow r \Leftrightarrow p \rightarrow (q \rightarrow r)$$

Exportation

$$(p \rightarrow q) \wedge (p \rightarrow \neg q) \Leftrightarrow \neg p$$

Absurdity

$$p \rightarrow q \Leftrightarrow \neg q \rightarrow \neg p$$

Contrapositive

$$p \wedge (p \vee q) \Leftrightarrow p$$

Absorption law

$$p \vee (p \wedge q) \Leftrightarrow p$$

Absorption law

Logical Equivalences



a) Truth table:

a) Truth Table (for $(p \rightarrow q) \wedge (p \rightarrow \neg q) \Leftrightarrow \neg p$)

p	$\neg p$	q	$\neg q$	$p \rightarrow q$	$p \rightarrow \neg q$	$p \rightarrow q \wedge p \rightarrow \neg q$
T	F	T	F	T	F	F
T	F	F	T	F	T	F
F	T	T	F	T	T	T
F	T	F	T	T	T	T

Logical Equivalences



b) Symbolic Derivation:

Show that $\neg(p \rightarrow q)$ and $p \wedge \neg q$ are logically equivalent.

solution:

$$\neg(p \rightarrow q) \equiv \neg(\neg p \vee q) \text{ by implication law}$$

$$\equiv \neg(\neg p) \wedge \neg q \text{ by the second De Morgan law}$$

$$\equiv p \wedge \neg q \text{ by the double negation law}$$



Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.

$$\begin{aligned}\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) \text{ by the second De Morgan law} \\ &\equiv \neg p \wedge [\neg(\neg p) \vee \neg q] \text{ by the first De Morgan law} \\ &\equiv \neg p \wedge (p \vee \neg q) \text{ by the double negation law} \\ &\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) \text{ by the second distributive law} \\ &\equiv F \vee (\neg p \wedge \neg q) \quad \text{because } \neg p \wedge p \equiv F \\ &\equiv (\neg p \wedge \neg q) \vee F \quad \text{by the commutative law for disjunction} \\ &\equiv \neg p \wedge \neg q \quad \text{by the identity law for } F\end{aligned}$$



Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

Solution:

To show that this statement is a tautology, we will use logical equivalences to demonstrate that it is logically equivalent to T.

$$\begin{aligned}(p \wedge q) \rightarrow (p \vee q) &\equiv \neg(p \wedge q) \vee (p \vee q) \text{ by implication law} \\ &\equiv (\neg p \vee \neg q) \vee (p \vee q) \text{ by the first De Morgan law} \\ &\equiv (\neg p \vee p) \vee (\neg q \vee q) \text{ by the associative and} \\ &\hspace{15em} \text{commutative laws for disjunction} \\ &\equiv T \vee T \text{ by trivial tautology and the commutative law for} \\ &\hspace{15em} \text{disjunction} \\ &\equiv T \hspace{15em} \text{by the domination law}\end{aligned}$$

Predicate logic.



- The area of logic which deals with predicate is known as predicate logic.
- A predicate is a condition that a variable needs to satisfy.
- Predicate are propositions containing variables.

Lets take a statement “ $x > 5$ ” is this statement a proposition? The answer is no. Whenever the statements have variable(s) in them we cannot say those statements as a proposition.

The question here is can we make such statements to propositions? The answer here is yes.

- In the above statement there are two parts one is the variable part called “subject” and another is relation part “ >5 ” called “predicate”.

Predicate logic.



- We can denote the statement “ $x > 5$ ” by $P(x)$ where P is predicate “ >5 ” and x is the variable.
- We also call P as a propositional function where $P(x)$ gives value of P at x .
- Once value is assigned to the propositional function then we can tell whether it is true or false i.e. a proposition.

For e.g. if we put the value of x as 3 and 7 then we can conclude that $P(3)$ is false since 3 is not greater than 5 and $p(7)$ is true since 7 is greater than 5.

Predicate logic.



Example: Let $Q(x, y)$ denote the statement “ $x = y + 3$.” What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?

Solution:

To obtain $Q(1, 2)$, set $x = 1$ and $y = 2$ in the statement $Q(x, y)$. Hence, $Q(1, 2)$ is the statement “ $1 = 2 + 3$,” which is false. The statement $Q(3, 0)$ is the proposition “ $3 = 0 + 3$,” which is true.

Quantifiers



- Quantifiers are the tools to make the propositional function a proposition.
- Construction of proposition from the predicates using quantifier is called quantification.
- The variables that appear in the statement can take different possible values and all the possible values that the variable can take forms a domain called “Universe of Discourse” or “Universal set”.
- We study two types of quantifier
 - 1) Universal quantifier
 - 2) Existential quantifier.

Quantifiers



1) Universal Quantifier:

- It states that the statements within its scope are true for every value of the specific variable.
- It is denoted by the symbol $\forall$.
- The process of converting predicate into proposition using universal quantifier is called universal quantification.
- The universal quantification of $P(x)$ is the statement “ $P(x)$ for all values of x in the domain.”

The notation $\forall x P(x)$ denotes the universal quantification of $P(x)$. Here $\forall$ is called the universal quantifier. We read $\forall x P(x)$ as “for all x $P(x)$ ” or “for every x $P(x)$.”

- An element for which $P(x)$ is false is called a counterexample of $\forall x P(x)$.

Quantifiers



Example:

Let $P(x)$ be the statement “ $x + 1 > x$.” What is the truth value of the quantification $\forall x P(x)$, where the domain consists of all real numbers?

Solution:

Because $P(x)$ is true for all real numbers x , the quantification $\forall x P(x)$ is true.

Quantifiers



Example:

Let $Q(x)$ be the statement “ $x \leq 2$.” What is the truth value of the quantification $\forall x Q(x)$, where the domain consists of all real numbers?

Solution:

$Q(x)$ is not true for every real number x , because, for instance, $Q(3)$ is false. That is, $x = 3$ is a counterexample for the statement $\forall x Q(x)$. Thus $\forall x Q(x)$ is false.

Quantifiers



2) Existential Quantifier:

- It state that the statements within its scope are true for some value of the specific variable .
- It is denoted by $\exists$.
- The existential quantification of $P(x)$, denoted by $\exists x P(x)$, is a proposition “ $P(x)$ is true for some values of x in the universe of discourse”.
- The other forms of representation include “there exists x such that $P(x)$ is true” or “ $P(x)$ is true for at least one x ”.

Quantifiers



Example:

Let $P(x)$ denote the statement “ $x > 3$.” What is the truth value of the quantification $\exists x P(x)$, where the domain consists of all real numbers?

Solution:

Because “ $x > 3$ ” is sometimes true—for instance, when $x = 4$ the existential quantification of $P(x)$, which is $\exists x P(x)$, is true.

Quantifiers



Example:

Let $Q(x)$ denote the statement “ $x = x + 1$.” What is the truth value of the quantification $\exists x Q(x)$, where the domain consists of all real numbers?

Solution:

Because $Q(x)$ is false for every real number x , the existential quantification of $Q(x)$, which is $\exists x Q(x)$, is false.

Quantifiers



Precedence of Quantifiers:

- The quantifiers $\forall$ and $\exists$ have higher precedence than all logical operators from propositional calculus. For example, $\forall x P(x) \vee Q(x)$ is the disjunction of $\forall x P(x)$ and $Q(x)$.
- In other words, it means $(\forall x P(x)) \vee Q(x)$ rather than $\forall x(P(x) \vee Q(x))$.

Binding Variables:

- When a quantifier is used on the variable, we say that this occurrence of the variable is **bound**.
- An occurrence of a variable that is not bound by a quantifier is said to be **free**.

Eg: In the statement $\exists x(x + y = 1)$, the variable x is bound by the existential quantification $\exists x$, but the variable y is free because it is not bound by a quantifier and no value is assigned to this variable. This illustrates that in the statement $\exists x(x + y = 1)$, x is bound, but y is free.

Quantifiers



Scope: The part of a logical expression to which a quantifier is applied is called the scope of this quantifier. We use parenthesis to give scope of the quantifier.

Example:

In the statement $\exists x(P(x) \wedge Q(x)) \vee \forall x R(x)$, *all variables are bound. The scope of the first quantifier, $\exists x$, is the expression $P(x) \wedge Q(x)$ because $\exists x$ is applied only to $P(x) \wedge Q(x)$, and not to the rest of the statement.*

Similarly, the scope of the second quantifier, $\forall x$, *is the expression $R(x)$. That is, the existential quantifier binds the variable x in $P(x) \wedge Q(x)$ and the universal quantifier $\forall x$ binds the variable x in $R(x)$.*



Nested quantifier:

- If we use a quantifier that appears within the scope of another quantifier then it is called nested quantifier.

Example:

$\forall x \exists y (x + y = 0)$ is the same thing as $\forall x Q(x)$, where $Q(x)$ is $\exists y P(x, y)$, where $P(x, y)$ is $x + y = 0$.



Negating Quantified Expressions:

- We will often want to consider the negation of a quantified expression. For instance, consider the statement “Every student in your class has taken a course in calculus.”
- This statement is a universal quantification, namely, $\forall x P(x)$, where $P(x)$ is the statement “ x has taken a course in calculus” and the domain consists of the students in your class. The negation of this statement is “It is not the case that every student in your class has taken a course in calculus.”
- This is equivalent to “There is a student in your class who has not taken a course in calculus.” And this is simply the existential quantification of the negation of the original propositional function, namely,

$$\exists x \neg P(x).$$

This example illustrates the following logical equivalence:

$$\neg \forall x P(x) \equiv \exists x \neg P(x).$$

Translating the Sentences into Logical Expression



Example 1

Translate “not every integer is even” where the universe of discourse is set of integers.

Solution:

Let $E(x)$ denotes x is even.

$$\neg \forall x E(x)$$

Translating the Sentences into Logical Expression



Express the statements “Some student in this class has visited Mexico” and “Every student in this class has visited either Canada or Mexico” using predicates and quantifiers.

Solution:

The statement “Some student in this class has visited Mexico” means that “There is a student in this class with the property that the student has visited Mexico.”

We can introduce a variable x , so that our statement becomes “There is a student x in this class having the property that x has visited Mexico.”

We introduce $M(x)$, which is the statement “ x has visited Mexico.” If the domain for x consists of the students in this class, we can translate this first statement as $\exists x M(x)$.

Translating the Sentences into Logical Expression



However, if we are interested in people other than those in this class, we look at the statement a little differently. Our statement can be expressed as

“There is a person x *having the properties that x is a student in this class and x has visited Mexico.*”

- In this case, the domain for the variable x *consists of all people. We introduce $S(x)$ to represent “ x is a student in this class.” $M(x)$, which is the statement “ x has visited Mexico.”*
- *Our solution becomes $\exists x(S(x) \wedge M(x))$*

Translating the Sentences into Logical Expression



Similarly, the second statement can be expressed as

“For every x in this class, x has the property that x has visited Mexico or x has visited Canada.”

let $C(x)$ be “ x has visited Canada.”

$M(x)$, be “ x has visited Mexico.”

$$\forall x(C(x) \vee M(x)).$$

Translating the Sentences into Logical Expression



However, if the domain for x consists of all people, our statement can be expressed as

“For every person x , if x is a student in this class, then x has visited Mexico or x has visited Canada.”

In this case, the statement can be expressed as

$$\forall x(S(x) \rightarrow (C(x) \vee M(x))).$$

Mathematical Reasoning



Rules of Reasoning:

- To draw conclusion from the given premise we must be able to apply some well defined steps that helps reaching the conclusion.
- These steps of reaching the conclusion are provided by rule of inference.

Rule 1: modus ponens(law of detachment):

$$\frac{p \rightarrow q}{p} \therefore q$$

- when both $p \rightarrow q$ and p are true, we know that q must also be true.
- i.e $(p \wedge (p \rightarrow q)) \rightarrow q$ is tautology.



Example:

Suppose that the conditional statement “If it snows today, then we will go skiing” and its hypothesis, “It is snowing today,” are true.

Then, by modus ponens, it follows that the conclusion of the conditional statement, “We will go skiing,” is true.

TABLE 1 Rules of Inference.

<i>Rule of Inference</i>	<i>Tautology</i>	<i>Name</i>
$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$	$p \rightarrow (p \vee q)$	Addition
$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$	$(p \wedge q) \rightarrow p$	Simplification
$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution



Example:

State which rule of inference is the basis of the following argument: “It is below freezing now . Therefore, it is either below freezing or raining now.”

Solution: Let p be the proposition “It is below freezing now” and q the proposition “It is raining now.” Then this argument is of the form

$$\frac{p}{\therefore p \vee q}$$

- This is an argument that uses the addition rule.



Example :

State which rule of inference is the basis of the following argument: “It is below freezing and raining now. Therefore, it is below freezing now.”

Solution:

Let p be the proposition “It is below freezing now,” and let q be the proposition “It is raining now.” This argument is of the form

$$\frac{p \wedge q}{\therefore p}$$

- This argument uses the simplification rule.

Using Rules of Inference to Build Arguments



- When there are many premises, several rules of inference are often needed to show that an argument is valid.
- The steps of arguments are displayed on separate lines, with the reason for each step explicitly stated.

Example:

Show that the premises “It is not sunny this afternoon and it is colder than yesterday,” “We will go swimming only if it is sunny,” “If we do not go swimming, then we will take a canoe trip,” and “If we take a canoe trip, then we will be home by sunset” lead to the conclusion “We will be home by sunset.”

Using Rules of Inference to Build Arguments



Solution:

Let p be the proposition “It is sunny this afternoon,” q the proposition “It is colder than yesterday,” r the proposition “We will go swimming,” s the proposition “We will take a canoe trip,” and t the proposition “We will be home by sunset.”

Then the premises become

- *$\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s$, and $s \rightarrow t$. The conclusion is simply t .*
- *We need to give a valid argument with premises $\neg p \wedge q, r \rightarrow p, \neg r \rightarrow s$, and $s \rightarrow t$ and conclusion t .*
- *We construct an argument to show that our premises lead to the desired conclusion as follows.*

Using Rules of Inference to Build Arguments



Step	Reason
1. $\neg p \wedge q$	<i>Premise/ Hypothesis</i>
2. $\neg p$	<i>Simplification using (1)</i>
3. $r \rightarrow p$	<i>Premise / Hypothesis</i>
4. $\neg r$	<i>Modus tollens using (2) and (3)</i>
5. $\neg r \rightarrow s$	<i>Premise</i>
6. s	<i>Modus ponens using (4) and (5)</i>
7. $s \rightarrow t$	<i>Premise</i>
8. t	<i>Modus ponens using (6) and (7)</i>

Using Rules of Inference to Build Arguments



Example:

Show that the premises “If you send me an e-mail message, then I will finish writing the program,” “If you do not send me an e-mail message, then I will go to sleep early,” and “If I go to sleep early, then I will wake up feeling refreshed” lead to the conclusion “If I do not finish writing the program, then I will wake up feeling refreshed.”

Solution:

Let p be the proposition “You send me an e-mail message,” q the proposition “I will finish writing the program,” r the proposition “I will go to sleep early,” and s the proposition “I will wake up feeling refreshed.” Then the premises are $p \rightarrow q$, $\neg p \rightarrow r$, and $r \rightarrow s$. The desired conclusion is $\neg q \rightarrow s$. We need to give a valid argument with premises $p \rightarrow q$, $\neg p \rightarrow r$, and $r \rightarrow s$ and conclusion $\neg q \rightarrow s$.

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This argument form shows that the premises lead to the desired conclusion.

Step	Reason
1. $p \rightarrow q$	<i>Premise</i>
2. $\neg q \rightarrow \neg p$	<i>Contra-positive of (1)</i>
3. $\neg p \rightarrow r$	<i>Premise</i>
4. $\neg q \rightarrow r$	<i>Hypothetical syllogism using (2) and (3)</i>
5. $r \rightarrow s$	<i>Premise</i>
6. $\neg q \rightarrow s$	<i>Hypothetical syllogism using (4) and (5)</i>

Rules of Inference for Quantified Statements



1) Universal instantiation: it is the rule of inference used to conclude that $P(c)$ is true, where c is a particular member of the domain, given the premise $\forall x P(x)$.

Universal instantiation is used when we conclude from the statement “All women are wise” that “Lisa is wise,” where Lisa is a member of the domain of all women.

2) Universal generalization: it is the rule of inference that states that $\forall x P(x)$ is true, given the premise that $P(c)$ is true for all elements c in the domain.

Universal generalization is used when we show that $\forall x P(x)$ is true by taking an arbitrary element c from the domain and showing that $P(c)$ is true.

Rules of Inference for Quantified Statements



3) Existential instantiation: it is the rule that allows us to conclude that there is an element c in the domain for which $P(c)$ is true if we know that $\exists x P(x)$ is true.

4) Existential generalization: it is the rule of inference that is used to conclude that $\exists x P(x)$ is true when a particular element c with $P(c)$ true is known.

That is, if we know one element c in the domain for which $P(c)$ is true, then we know that $\exists x P(x)$ is true.

Rules of Inference for Quantified Statements



TABLE 2 Rules of Inference for Quantified Statements.

<i>Rule of Inference</i>	<i>Name</i>
$\frac{\forall x P(x)}{\therefore P(c)}$	Universal instantiation
$\frac{P(c) \text{ for an arbitrary } c}{\therefore \forall x P(x)}$	Universal generalization
$\frac{\exists x P(x)}{\therefore P(c) \text{ for some element } c}$	Existential instantiation
$\frac{P(c) \text{ for some element } c}{\therefore \exists x P(x)}$	Existential generalization



Example:

Show that the premises “Everyone in this discrete mathematics class has taken a course in computer science” and “Marla is a student in this class” imply the conclusion “Marla has taken a course in computer science.”

Solution:

Let $D(x)$ denote “ x is in this discrete mathematics class,” and let $C(x)$ denote “ x has taken a course in computer science.” Then the premises are

$\forall x(D(x) \rightarrow C(x))$ and $D(\text{Marla})$. The conclusion is $C(\text{Marla})$.

The following steps can be used to establish the conclusion from the premises.

Step

1. $\forall x(D(x) \rightarrow C(x))$

2. $D(\text{Marla}) \rightarrow C(\text{Marla})$

3. $D(\text{Marla})$

4. $C(\text{Marla})$

Reason

Premise

Universal instantiation from (1)

Premise

Modus ponens from (2) and (3)

Show that the premises “A student in this class has not read the book,” and “Everyone in this class passed the first exam” imply the conclusion “Someone who passed the first exam has not read the book.”

Solution: Let $C(x)$ be “ x is in this class,” $B(x)$ be “ x has read the book,” and $P(x)$ be “ x passed the first exam.” The premises are $\exists x(C(x) \wedge \neg B(x))$ and $\forall x(C(x) \rightarrow P(x))$. The conclusion is $\exists x(P(x) \wedge \neg B(x))$. These steps can be used to establish the conclusion from the premises.

Step

Reason

- | | |
|---------------------------------------|-------------------------------------|
| 1. $\exists x(C(x) \wedge \neg B(x))$ | Premise |
| 2. $C(a) \wedge \neg B(a)$ | Existential instantiation from (1) |
| 3. $C(a)$ | Simplification from (2) |
| 4. $\forall x(C(x) \rightarrow P(x))$ | Premise |
| 5. $C(a) \rightarrow P(a)$ | Universal instantiation from (4) |
| 6. $P(a)$ | Modus ponens from (3) and (5) |
| 7. $\neg B(a)$ | Simplification from (2) |
| 8. $P(a) \wedge \neg B(a)$ | Conjunction from (6) and (7) |
| 9. $\exists x(P(x) \wedge \neg B(x))$ | Existential generalization from (8) |



Methods of Proving



1) **Direct Proofs:**

- In a direct proof, we assume that p is *true* and use *axioms*, definitions, and previously proven theorems, together with rules of inference, to show that q must also be true.
- It is quite straightforward, with a sequence of steps leading from the hypothesis to the conclusion.

Direct Proofs:



Give a direct proof of the theorem “If n is an odd integer, then n^2 is odd.”

Solution:

To begin a direct proof of this theorem, we assume that the hypothesis of this conditional statement is true, namely, we assume that n is odd.

By the definition of an odd integer, it follows that

$$n = 2k + 1, \text{ where } k \text{ is some integer.}$$

We want to show that n^2 is also odd.

We can square both sides of the equation $n = 2k + 1$ to obtain a new equation that expresses n^2 .

$$\begin{aligned} n^2 &= (2k + 1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1. \end{aligned}$$

By the definition of an odd integer, we can conclude that n^2 is an odd integer (it is one more than twice an integer).

Consequently, we have proved that if n is an odd integer, then n^2 is an odd integer.

Eg: Give a direct proof that if m and n are both perfect squares, then nm is also a perfect square.

Solution:

To produce a direct proof of this theorem, we assume that the hypothesis of this conditional statement is true, namely, we assume that m and n are both perfect squares.

By the definition of a perfect square, it follows that there are integers s and t such that $m = s^2$ and $n = t^2$.

The goal of the proof is to show that mn must also be a perfect square when m and n are; looking ahead we see how we can show this by substituting s^2 for m and t^2 for n into mn .

$$\begin{aligned} mn &= s^2 t^2. \\ &= s^2 t^2 \\ &= (ss)(tt) \\ &= (st)(st) = (st)^2 \end{aligned}$$

By the definition of perfect square, it follows that mn is also a perfect square, because it is the square of st , which is an integer.

We have proved that if m and n are both perfect squares, then mn is also a perfect square.

2. Indirect proofs



- Proofs of theorems of this type that are not direct proofs, that is, it do not start with the premises and end with the conclusion.
- It is also known as proof by contraposition.
- Proofs by contraposition make use of the fact that the conditional statement $p \rightarrow q$ is equivalent to its contra-positive, $\neg q \rightarrow \neg p$.
- This means that the conditional statement $p \rightarrow q$ can be proved by showing that its contra-positive, $\neg q \rightarrow \neg p$, is true.
- In a proof by contraposition of $p \rightarrow q$, we take $\neg q$ as a premise, and using axioms, definitions, and previously proven theorems, together with rules of inference, we show that $\neg p$ must follow.

Example: Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

solution:

The first step in a proof by contraposition is to assume that the conclusion of the conditional statement

“If $3n + 2$ is odd, then n is odd” is false; namely, assume that n is even. Then, by the definition of an even integer, $n = 2k$ for some integer k . Substituting $2k$ for n , we find that

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1).$$

This tells us that $3n + 2$ is even (because it is a multiple of 2), and therefore not odd.

This is the negation of the premise of the theorem.

Because the negation of the conclusion of the conditional statement implies that the hypothesis is false, the original conditional statement is true.

Our proof by contraposition succeeded;

We have proved the theorem “If $3n + 2$ is odd, then n is odd.”

Indirect method

Prove that if n is an integer and n^2 is odd, then n is odd.

Solution:

Prove by contraposition. We take as our hypothesis the statement that n is not odd. This means that n is even.

This implies that there exists an integer k such that $n = 2k$.

To prove the theorem,

we need to show that this hypothesis implies the conclusion that n^2 is not odd, i.e., that n^2 is even.

Can we use the equation $n = 2k$ to achieve this?

By squaring both sides of this equation, we obtain

$$n^2 = 4k^2 = 2(2k^2),$$

which implies that n^2 is also even because $n^2 = 2t$, where $t = 2k^2$.

We have proved that if n is an integer and n^2 is odd, then n is odd.

Our attempt to find a proof by contraposition succeeded.

3.Proofs by Contradiction



- The steps in proof of implication $p \rightarrow q$ by contradiction are:
 - assume $(p \wedge \neg q)$ is true.
 - try to show the above assumption is false.
 - If the assumption is found to be false then $p \rightarrow q$ is true.

Give a proof by contradiction of the theorem “If $3n + 2$ is odd, then n is odd.”

Solution:

Let p be “ $3n + 2$ is odd” and q be “ n is odd.” To construct a proof by contradiction, assume that both p and $\neg q$ are true. That is, assume that $3n + 2$ is odd and n is not odd.

Because n is not odd, we know that it is even. Because n is even, there is an integer k such that $n = 2k$.

This implies that,

$$\begin{aligned} & 3n + 2 \\ &= 3(2k) + 2 \\ &= 6k + 2 \\ &= 2(3k + 1). \\ &= 2t \qquad \text{where } t = 3k + 1, \end{aligned}$$

$3n + 2$ is even.

So by contradiction, we can prove that if $3n + 2$ is odd, then n is odd.

If a^2 is an even number, then a is an even number.

Solution:

Assume ' a ' is an odd number and a^2 is an even number.

$$a=2k+1 \quad (\text{by definition of odd integers})$$

Then,

$$\begin{aligned} a^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \\ &= 2m + 1 \end{aligned}$$

where $m = 2k^2 + 2k$, is any integer

a^2 is odd

So by contradiction, we can prove that

If a is an even number, then a^2 is an even number.

4.Proof by Cases



Sometimes we cannot prove a theorem using a single argument that holds for all possible cases.

- We now introduce a method that can be used to prove a theorem, by considering different cases separately.
- To prove a conditional statement of the form

$$(p1 \vee p2 \vee \cdots \vee pn) \rightarrow q$$

Can prove by $[(p1 \rightarrow q) \wedge (p2 \rightarrow q) \wedge \cdots \wedge (pn \rightarrow q)]$

Prove that $(n + 1)^3 \geq 3n$ if n is a positive integer with $n \leq 4$.

Solution:

We use a proof by cases. We only need verify the inequality $(n + 1)^3 \geq 3n$ when $n = 1, 2, 3$, and 4 .

For $n = 1$,

*we have $(n + 1)^3 = 2^3 = 8$ and $3n = 3*1 = 3$;*

For $n = 2$,

*we have $(n + 1)^3 = 3^3 = 27$ and $3n = 3*2 = 9$;*

For $n = 3$,

*we have $(n + 1)^3 = 4^3 = 64$ and $3n = 3*3 = 27$; and*

For $n = 4$, *we have $(n + 1)^3 = 5^3 = 125$ and $3n = 3*4 = 12$.*

In each of these four cases, we see that $(n + 1)^3 \geq 3n$. We have used the method of cases to prove that $(n + 1)^3 \geq 3n$ if n is a positive integer with $n \leq 4$.

Prove that if $n \in \mathbb{Z}$, then $n^2 + 3n + 4$ is even.

Solution: $n \in \mathbb{Z}$ (integer) then there will be 2 case either n is odd or even.

case 1: n is odd

since n is odd n will be equal to $2x+1$

Now,

substitute the value of n in conclusion

$$\begin{aligned}n^2 + 3n + 4 &= (2x + 1)^2 + 3(2x + 1) + 4 \\&= 4x^2 + 4x + 1 + 6x + 3 + 4 \\&= 4x^2 + 10x + 8 \\&= 2(2x^2 + 5x + 4) \\&= 2t\end{aligned}$$

By the definition of even integer it will be even.

case 2: n is even

since n is even n will be equal to $2x$

Now,

substitute the value of n to the conclusion

$$\begin{aligned}n^2 + 3n + 4 &= (2x)^2 + 3(2x) + 4 \\&= 2(2x^2 + 3x + 2)\end{aligned}$$

Which is also even.

So in both cases the conclusion is even i.e. true